

Behavioural Data as Subject-Owned Capital: Five Predicates and the Architecture that Prices Them

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Working paper, revision v2. Prepared for WEIS 2027 (Thrust T3, Value and Governance). Format-free venue; written for the security economist and the economist of data markets, not for the cryptographer. The one information-theoretic result the argument rests on is imported from a companion technical paper (WP-07) and cited, never re-proven here; the durability leg of the thesis is gated on that companion being a real, citable paper at submission (Section 6).

Abstract

Behavioural data, the record of what a person attends to, asks, and declines to do, is treated here as subject-owned capital: a stock satisfying the five defining predicates of a capital form (it yields returns, compounds through reputation as a conjecture, can be invested in, gates opportunity access, and transfers across contexts). The framing arose by analogy with the six-capital integrated-reporting frame, from which the informal “seventh capital” label derives; that ordinal is provenance only and the argument rests no weight on it. Against both the incumbent treatment of the stock as firm-owned capital and the data-as-labour proposal that would price it as a wage, we hold that it is capital the data subject owns. This is a propertization claim, and we engage the critical propertization literature directly, answering its two load-bearing objections (alienability and thin markets) by architecture rather than doctrine. The subject nonetheless captures a near-zero share of the surplus the stock generates. We argue this share is the subgame-perfect equilibrium of one bargaining game whose rules are the consent interface: notice-and-consent is a single-round take-it-or-leave-it offer whose subject share equals her near-zero outside option (exclusion from the service), while a bilateral propose-and-respond protocol (IEEE Std 7012-2025) inverts the offer direction and moves the interior share to the subject’s bargaining power. We replace two magnitudes circulating in the project’s prior essays, which we decline to reproduce, with a valuation carrying every assumption in the same table: a bounded surplus-appropriation share, and a non-identified observer-to-broker realised-value gap that the returns-to-scale evidence attributes to market-position rent rather than to any super-additivity of aggregated data. We report that rent finding at the prominence of a positive result: raw behavioural-data aggregation does not compound, so a subject-side aggregator redistributes a positional rent rather than unlocking latent value. Finally we make precise the sense in which the stock is durable: it rests on an architectural separation of an observing boundary agent S from an acting delegation agent M, whose guarantee is information-theoretic, imported from the

companion paper, and time-indexed. It holds against a fixed adversary class under non-collusion and, where the declared capacity-deficit condition holds, its reconstruction residual $R(t)$ is bounded below one at evaluation and rises as the adversary’s background information accumulates against the fixed archive, so the asset is durable only for a finite, adversary-relative horizon. The sovereignty the architecture protects therefore has a market price, currently paid to someone else. We report the negative results, the unidentified rent, the unmeasured appreciation claim, the open Sybil-resistance conjecture, and the shadow-exchange hazard to the separation’s incentive compatibility, with the same prominence as the positive ones.

1. Introduction and contributions

Consider the position of an implementer building a data market, not that of a framework author. Two facts about deployed personal-data markets are hard to dispute. First, behavioural data is valuable: firms that aggregate and model it earn returns on it. Second, the person whose behaviour produced the data captures almost none of that value. The standard readings of the second fact are that the subject does not care about privacy (contradicted by stated preferences), or that the data is worth little at the margin (true of an atomised record, false of the aggregate), or that the subject has agreed to the arrangement (true only in the thin sense that she clicked accept).

This paper takes a mechanism view of the second fact. The subject’s share of the surplus is the equilibrium of a bargaining game whose rules are set by the consent interface. Read as the extensive form of an alternating-offers game, notice-and-consent is an ultimatum game in which the observing counterparty is the sole proposer, and its unique subgame-perfect outcome hands the subject only her outside option, near zero because she is not admitted to the venue where behavioural data clears. That a sole proposer captures almost all the surplus in a one-shot take-it-or-leave-it game is the textbook ultimatum result (Guth et al. 1982; the one-shot limit of Rubinstein 1982 alternating offers), and formal data-market models in which an intermediary’s bargaining power fixes the consumer’s residual surplus reach the same place (Gu 2024); we claim neither the derivation nor its direction as new. The near-zero share is not evidence that the data is worthless; it is the predicted outcome of the interface. Change the interface and the prediction changes. What the paper takes from this is not the equilibrium but its application: the subject-side share is written as a structural venue-exclusion parameter rather than as a taste for privacy, and the reform is a specific permutation of the proposer role.

The change we study is a bilateral propose-and-respond protocol in which the subject proposes terms and the counterparty accepts, negotiates once, or declines. This structure is a published standard, IEEE Std 7012-2025, Machine Readable Personal Privacy Terms; that inverting the proposer to shift bargaining power is the standard’s own stated purpose. It inverts the direction of the offer and, when paired with a credible outside option, moves the subject’s equilibrium share off the floor and into an interior range governed by bargaining power rather than by appropriation.

For that repricing to be worth anything, the stock being priced must be defensible: the value of a subject’s sovereignty over her behavioural record collapses if an observer can reconstruct the record anyway. We therefore rest the economics on an architectural claim, that an observing boundary agent S and an acting delegation agent M can be structurally separated so that S never accumulates the context M holds, with structural context erasure between agent invocations. The information-theoretic content of that separation, a bound on what S can reconstruct, is developed and proved in

a companion paper (WP-07), imported and cited, never re-proven. What matters for the economics is stated with its preconditions (Section 3, Assumptions A1 to A3): the guarantee holds against a fixed adversary class under non-collusion, and where a declared capacity-deficit condition holds its reconstruction residual $R(t)$ is bounded below one at evaluation. $R(t)$ is not a static ceiling; it is an erosion clock. The archive and the disclosed budgets are fixed, and the residual rises because the adversary's background information, its linkage corpus and side priors, accumulates along calendar time and shrinks the record's residual uncertainty. The companion paper proves the erosion's form and direction for a declared background model; the rate of erosion is a register conjecture, not a theorem. The shelf life is therefore finite and what expires is the informed deficit condition, not the architecture. A guarantee with a finite, adversary-relative horizon is exactly what makes the sovereignty an asset durable for a term rather than forever.

Each of the contributions below was put to an adversarial prior-art search, and each returned the same verdict: the broad object is already owned by named prior work, and only a narrow residue survives. What the paper defends as novel is therefore not any single object but the conjunction of those residues held within one instrument, and it is offered as a candidate that survived a named search rather than as an empty cell. Two residues are load-bearing. The first is the pricing of a finite, adversary-relative, time-drifting reconstruction bound $R(t)$ as inalienable subject-owned capital, with the observer-to-subject value gap decomposed as market-position rent; this sits beside, and must be distinguished from, the differential-privacy-markets literature that already prices a disclosure budget as a commodity. That budget is not adversary-independent: it is calibrated against a worst-case adversary and accounted along a query and release-composition clock, whereas $R(t)$ indexes a fixed release to an adversary's informational capability growing along the calendar horizon, a distinct clock we develop and scope in Section 2. The second is a directed internal falsifier of the data-as-labour remedy: in the settings the returns-to-scale evidence has measured, the marginal product of an additional raw record is approximately zero at operating scale, so raising the subject's compensation toward that marginal product self-defeats there, and the redistributive object is re-specified as venue and clearing-position rent rather than a wage; the falsifier's scope is exactly the scope of that evidence, stated in Section 5 (Sections 2 and 5).

Contributions. This paper contributes:

1. A capital-class formalisation of behavioural data, with its five properties (returns, reputation-scoped compounding, investability, opportunity access, cross-context transfer) restated as measurable predicates rather than metaphors, and positioned within the critical propertisation literature rather than against it (Sections 2 and 3).
2. A valuation methodology that replaces two magnitudes from the project's prior essay lineage, which we do not reproduce as measured facts, with a bounded surplus-appropriation share and a non-identified observer-to-broker realised-value gap that the returns-to-scale evidence attributes to market-position rent rather than to super-additive aggregation; the same evidence supplies a directed correction to the data-as-labour programme, since a marginal product near zero for the raw record leaves no wage to raise, with every assumption carried in the same table as the quantity it feeds and all sensitivity expressed as ranges (Section 5).
3. A market mechanism, consent-interface bargaining for price discovery, a subject-side aggregator for rent redistribution, and a non-transferable quality stake for signalling, through which the sovereignty value is realised for the subject or lost to the observer (Sections 4 and 5).
4. A statement of the architectural guarantee as imported, time-indexed, and finite-horizon, together with the economic reading of its horizon: the asset is durable only for a term, and only if

the companion paper that proves the guarantee is a citable companion at submission (Sections 3 and 6).

What this paper does not claim. The adversarial search that shaped the contributions above also fixed their ceiling, and the boundary is stated here with the same prominence as the claims. We do not claim to be first to treat personal data as an ownable asset: propertisation of personal data is a decades-old, and largely critical, literature (Section 2). We do not claim to be first to price a privacy or disclosure system property as a market quantity: the differential-privacy-markets strand prices the disclosure bound as a traded commodity and occupies that cell already (Ghosh and Roth 2011, and successors). We do not claim the derivation of the subject’s near-zero share as a subgame-perfect equilibrium; that is the textbook ultimatum result. We do not claim the finite, adversary-relative durability horizon as a new form; an adversary-relative shrinking horizon is the structure of Mosca’s cryptographic shelf-life inequality (Mosca 2018), from which our term structure descends only by substituting an information-theoretic reconstruction bound for the cryptographic break-time clock. And we do not claim the diagnosis that intermediaries capture the surplus as data aggregates; that diagnosis is formalised elsewhere (Bergemann, Bonatti and Gan 2022). Each of these broad objects is prior art; what is defended is the narrow conjunction that sits on top of them.

No claim in this paper is strengthened to make the argument flow. Where a quantity is not identified we say so; where a directional claim is unmeasured we label it a conjecture and name the measurement that would settle it; and the one result that runs against the project’s own prior essays, that raw-data aggregation does not compound, is reported with the prominence of a confirmation.

2. Related work: positioning inside five literatures

The thesis sits at the intersection of five literatures, and its contribution is best seen as a specific move within each rather than a departure from all of them.

The economics of privacy. Acquisti, Taylor and Wagman (2016) establish that privacy is jointly cost- and benefit-bearing, context-dependent, and aggregation-dependent: the same records are worth little atomised and much in combination. Acquisti, John and Loewenstein (2013) measure the gap between what individuals will accept to disclose and what they will pay to protect. We treat their willingness-to-accept over willingness-to-pay endowment ratio as consistent evidence of the appropriation problem, not as its signature: the elicited value wedge is a small multiple, and a larger participation-proportion ratio in the same experiment is a ratio of subject shares under two framings, not of valuations, so we use it only as direction-only corroboration (Section 5.1). The subject values her own data above the price the market pays her, and the gap is structural.

Pricing privacy as a system property: the differential-privacy-markets strand. The paper prices a property of the release mechanism rather than a subject’s reported impact, so it must sit beside the literature that already prices such a property as a market quantity, and this is the strand the current framing most directly answers. Ghosh and Roth (2011) auction a differential-privacy budget, compensating subjects for a bounded disclosure and deriving an equilibrium price for it; Li et al. (2013) build a theory of pricing private-data queries; Fleischer and Lyu (2012) and Cummings et al. (2015) extend the pricing of privacy loss to correlated and to aggregated-estimation settings. This strand prices a system property, the disclosure bound, as a traded commodity, so the claim that no market attaches to a system property does not survive contact with it, and we do not make that claim. The move from a subject-reported impact assessment to a quantified system-level metric

is likewise not ours to claim: Wagner and Boiten (2018) already argue that the privacy-impact-assessment tradition should be replaced by quantified metrics, and we adopt the system-property unit rather than propose the transition to it. What the paper adds over this strand is a specific and narrower object, tested against it rather than asserted past it. The differential-privacy budget priced there is not adversary-independent: it is calibrated against a worst-case adversary of unbounded computation and arbitrary auxiliary knowledge, entering the guarantee as a supremum (Mironov 2017; Bun and Steinke 2016; Dong, Roth and Su 2022), and it carries a genuine temporal axis, with privacy loss accounted as it accumulates across composed mechanisms (Kairouz, Oh and Viswanath 2015; Dwork, Rothblum and Vadhan 2010) or across a stream of published outputs under continual observation (Dwork, Naor, Pitassi and Rothblum 2010); adversary-aware refinements parametrise which fixed adversary class the guarantee is calibrated against (Cummings et al. 2024; Swanberg et al. 2025). What that literature holds fixed is therefore not the adversary but the clock: its temporal axis is added queries or releases against a held-fixed worst-case adversary. $R(t)$ runs a different clock. It indexes a single fixed release, never re-queried, whose per-subject reconstruction residual rises as the adversary’s informational capability, its accumulating linkage corpus and side priors rather than its computational power, which differential privacy already saturates, grows along calendar time; and that residual is held as inalienable subject-owned capital that scoped disclosure does not transfer, with the observer-to-subject value gap decomposed as market-position rent rather than as super-additive aggregation. The companion paper carries this clock as proved material rather than as framing: conditioning on the accumulating background cannot raise the disclosed transcript’s leakage above its budget, while the reconstruction floor erodes through the record’s residual entropy given that background; the protection does not leak more, it matters less. Where differential privacy does run a wall clock, in age-dependent and temporally-discounted formulations and in the data-value decay of Valavi et al. (2022), the polarity is opposite: the guarantee strengthens as data ages through relevance decay. We state this as a clock distinction and not a categorical barrier, because adversary-aware differential privacy already carries adversary capability as an explicit static parameter and computational differential privacy already recognises informally that a fixed release erodes as adversary capability grows, so a calendar-indexed capability clock is a natural but as-yet unformalised extension rather than an impossibility. It is that quadruple, the calendar-time adversary-capability clock, structural inalienability, subject-side holding, and rent decomposition, and not the pricing of a system property as such, that the paper defends, and it defends it as the residue that survived a search against this strand, scoped to that strand.

Data as labour versus data as capital. Arrieta-Ibarra et al. (2018) and Posner and Weyl (2018) argue that behavioural data should be treated as labour, remunerated like a wage, against the incumbent treatment as firm-owned capital; Lanier (2013) frames the same intuition as data dignity. Our position is a deliberate third one, and the tension is the paper’s clearest point of contact with the literature. We agree the returns are misappropriated; we disagree on the remedy. The data-as-labour proposal keeps the stock in the firm’s hands and compensates the subject for a flow of contributions; ours reassigns ownership of the stock to the subject and treats her return as a return on owned capital. A wage is priced per contribution and stops when it stops; an owned, durable stock yields as long as it is held and can be invested, compounded (reputationally; Section 3), and transferred. The capital framing is the claim that behavioural data has the second character, provided an architecture keeps the stock from depreciating in the subject’s hands. That proviso is the whole of Sections 3 and 6. One point of contact here is sharper than a disagreement over remedy, and it is stated as a directed correction rather than a fresh claim. The data-as-labour remedy raises the subject’s compensation toward the marginal product of her contribution through a data-labour

cartel. The returns-to-scale evidence we rely on in Section 5 (Bajari et al. 2019 and the wider finding that data is not the new oil) indicates that the marginal product of an additional raw record is approximately zero at the scale where these markets operate, so there is no wage to raise and the remedy self-defeats on its own terms. The strength of this correction is bounded by the strength of that evidence, and we state the bound: the near-zero finding is measured in a small number of settings (retail demand forecasting in Bajari et al.; the estimation-task subsample argument in Varian; broker-profile accuracy in Neumann et al.), each with a specific production function, and it is an extrapolation, not a theorem, that every market where a data-labour cartel would bargain has the same diminishing shape. The falsifier is therefore directed, not universal: it binds wherever the root-N, flat-in-variety returns shape holds, and a measured market in which the marginal record retains substantial product would blunt it there and would be reported at the same prominence as the finding itself. We therefore re-specify the redistributive object not as a wage but as rent on venue access, and the instrument that redistributes it, a subject-side aggregator pooling bargaining position, is the data-union, data-cooperative and consortia programme’s device (Arrieta-Ibarra et al. 2018; Posner and Weyl 2018; Delacroix and Lawrence 2019), imported here rather than introduced. We offer this correction as a residue that survived a search against those same programmes, not as a claim that they are unaware of scale effects.

Propertisation of personal data, and why the architecture is not just a property right. The position that behavioural data is capital *the subject owns* is a propertisation claim, and propertisation of personal data has a decades-old and largely critical literature that the framing must answer rather than restate. Laudon (1996) proposed treating personal information as an owned, licensable asset traded in a national information market; Samuelson (2000) asked directly whether privacy could be protected by an intellectual-property-style right and was sceptical; Schwartz (2004) set out a model of propertised personal information but only with explicit safeguards, because he took the naive version to fail; Purtova (2015) and Prins (2006) analysed the European commodification debate and its hazards; and the data-trust and data-cooperative strand (Delacroix and Lawrence 2019) proposed fiduciary intermediation as an alternative to individual ownership altogether. Cofone (2022) carries the critique to the present: ownership proposals collapse into consent-based transfer rules, magnify the consent pathologies they were meant to cure, and cannot reach inferred data at all; the reconstruction-bound unit answers that last failure directly, because $R(t)$ prices what an observer can infer rather than what a subject transferred. A data-as-capital reading also exists in critical political economy, but there the capital is firm-held and extractive (Sadowski 2019); the position here relocates the stock to the subject, which is exactly the move the propertisation debate must vet. A referee who knows this literature will read subject-owned capital as a rebranding of propertisation unless the paper says what it adds. It adds two things, and both are structural rather than doctrinal.

The literature levels four standing objections at propertisation, and we take each in turn.

Objection (i): alienability defeats the protection. A right the subject can sell is a right she will be pressured to sell; under free alienability the protection is gone once the data transfers (Samuelson 2000). Schwartz’s (2004) response built *hybrid inalienability* into the right by legal construction. Our response makes the inalienability *structural*. What the subject discloses is a scoped, single-purpose disclosure, not a handover of the stock, because the observing side cannot reconstruct the underlying record from it: S and M are separated so S never accumulates the context M holds (structural context erasure between agent invocations), and the residual an observer can reconstruct is bounded by the guarantee imported from WP-07. That guarantee is not a static ceiling: it holds against a fixed adversary class (Assumption A2) under non-collusion (Assumption A1), and where the informed

capacity-deficit condition holds its residual $R(t)$ is bounded below one at evaluation and rises as the adversary's background information accumulates (Assumption A3). The shape of that horizon, an adversary-relative term that shortens as adversary capability grows, is not new: it is the structure of Mosca's cryptographic shelf-life inequality and the harvest-now-decrypt-later threat model (Mosca 2018), and what the paper substitutes is an information-theoretic reconstruction bound for that literature's cryptographic break-time clock, carried into a capital valuation (Sections 3 and 6). Within that term the asset cannot be fully alienated by disclosure, because disclosure does not transfer what it cannot reconstruct. Second, the quality stake (Section 4.3) is *non-transferable by construction*: it cannot be minted by fabricated accounts, purchased, or moved between accounts (register conjecture C42). This is Schwartz's inalienability as a property of the object rather than a clause a court must enforce.

Objection (ii): thin markets. Individual data is worth almost nothing atomised, transaction costs dwarf the per-record price, and no functioning market exists at the single-subject level. Our valuation confirms the premise and is sharper about its cause (Section 5): the atomised broker-resale price sits several orders of magnitude below the observer's realised lifetime capture, and the returns-to-scale evidence shows this gap is *not* super-additive aggregation the subject could never replicate. It is market-position rent, the discount a record suffers because the subject is not admitted to the venue where behavioural data clears (Sections 5.2, 5.3). The diagnosis that the data intermediary captures the total information value as the number of subjects grows, while each individual captures almost nothing because her data is informative about others through a social-data externality, is formalised by Bergemann, Bonatti and Gan (2022); we concede that this rent diagnosis is anticipated there rather than original to us, and locate the paper's residue in the directed falsifier of the labour remedy above and in carrying the diagnosis into the reconstruction-bound frame, not in the diagnosis itself. If the gap were compounding aggregation value, capturing it would require each subject to reproduce a platform's aggregation technology. Because it is positional rent, capturing it requires only pooling *bargaining position*, exactly the function of the subject-side aggregator, the mediator-of-individual-data, data-cooperative or data-trust instrument (Arrieta-Ibarra et al. 2018; Posner and Weyl 2018; Delacroix and Lawrence 2019). Two attributions keep this honest. The pooling form itself is anticipated inside the externality line, which proposes collective bargaining as the internalisation of the social-data externality (Bergemann and Wang 2021); what this paper contributes is the subject-side capital valuation the instrument operates on, not the instrument. And the formal externality lineage beyond Bergemann, Bonatti and Gan reinforces the diagnosis from three directions: informed consent still yields excessive collection under information externalities (Choi, Jeon and Kim 2019); correlation leakage depresses equilibrium data prices to the point where conditions exist under which closing the market improves welfare, a partial formal vindication of the ban position engaged under objection (iv) below (Acemoglu, Makhdoumi, Malekian and Ozdaglar 2022); and an intermediary's acquisition can harm consumers even when every individual sale is rational (Ichihashi 2021).

Objection (iii): distributional regressivity. Under alienable propertisation the poor sell privacy cheaply and the wealthy retain it. The structural inalienability of the reconstruction (objection (i)) means the regressive transaction, alienating the stock itself for a low price, is not the transaction the architecture supports; what is priced is scoped disclosure within a retained stock. The fiduciary form (Delacroix and Lawrence 2019) is designed to redistribute bargaining position toward those with the least of it. The honest limit: nothing in the architecture guarantees a non-regressive *distribution* of the rent it redistributes; it changes who is admitted to the venue, not how the surplus is split once inside, and that split is the projected sovereign-regime share s (Section 6). We claim

the architecture removes the specific regressive mechanism propertisation is charged with, forced alienation of the stock, not that it delivers a distributively just outcome.

Objection (iv): commodification is corrosive per se. Distinct from the three objections above, the anti-commodification position holds that pricing personal data is wrong in itself, not merely hazardous in execution. Radin (1987) grounds market-inalienability in personhood: some goods are constitutive of persons, and marketing them degrades both the good and the holder. Roessler (2015) applies the moral-limits-of-markets analysis to personal data trade directly and concludes against tradability; Véliz (2020; 2024) states the live form of the position, that the trade in personal data should be banned outright; and the position has doctrinal expression, with EU data-protection scholarship arguing the fundamental right is at odds with data trade as such (Custers and Malgieri 2022) and the European regulator holding that personal data “cannot be considered as a commodity” (EDPS Opinion 4/2017). Satz (2010) supplies the test form through her noxious-markets criteria: weak agency on the selling side, vulnerability, and harmful externalities on third parties. We answer this objection differently from the first three, by partial vindication rather than rebuttal. On the stock, the architecture agrees with the pole: the underlying record is never marketed, and its inalienability is structural rather than a doctrinal rule a legislature could repeal, which is the permanence the pole demands. In the Calabresi and Melamed (1972) taxonomy the entitlement is protected by an inalienability rule, with the rule enforced by a reconstruction bound rather than by a court. What the architecture prices is scoped disclosure within the retained stock, and on Satz’s own criteria the noxiousness of existing data markets traces to features the architecture removes: the seller’s weak agency is the ultimatum floor of Section 4, and the third-party externality is priced into the bound rather than scoped away, because $R(t)$ is defined against the adversary’s full inference capability, including inference from other subjects’ data, the end-run around anonymity and consent that Barocas and Nissenbaum (2014) identify. What we do not answer is the corruption claim in its strong form, that any pricing of any privacy-adjacent quantity degrades the good (Anderson 1993; Sandel 2012). Against that position the paper’s claim is only conditional: if a price is to exist at all, and the observed data economy prices behavioural data continuously without asking, then a bound-priced scoped disclosure within an inalienable stock is the pricing form that concedes most of what the objection defends.

A related normative strand attacks not the pricing but the individualist unit of analysis: data relations create population-level interests that no individual holder can represent (Viljoen 2021), privacy interests attach to groups and profiles as well as to persons (Taylor, Floridi and van der Sloot 2017), and individually rational disclosure undersupplies privacy as a public good (Fairfield and Engel 2015). The answer here is also partial. The social-data externality is carried inside the bound rather than left outside the transaction, and the remedy instrument is collective by construction, but the framework remains subject-indexed, and a fully relational governance of the population-level interest is more than a per-subject valuation can claim; we state this as a limit in Section 6.

Stated as the result of a search rather than as an assertion, the broad claims gathered here each have named prior owners, and were recorded as anticipated. That behavioural data can be owned was proposed and criticised long ago; that a disclosure system property can be priced as a market quantity is the differential-privacy-markets strand’s (Ghosh and Roth 2011, and successors), and that strand also already joins a priced formal quantity to records the buyer never receives, with arbitrage-freeness in query pricing acting as a bound on what combined purchases can reconstruct (Li et al. 2013) and federated designs extending the join to subject-side retention of the raw records (Zheng et al. 2021), so non-transfer alone is conceded as an established mechanism feature; that an

intermediary captures the surplus as data aggregates is Bergemann, Bonatti and Gan’s (2022); that an adversary-relative horizon shrinks a protection term is Mosca’s (2018). What survived the search is the narrow conjunction that sits across them: the two objections that sank naive propertisation are answered by structure rather than doctrine, with the inalienability theorised as a property of the capital stock, an inalienability rule in the Calabresi and Melamed (1972) sense enforced by the reconstruction bound, rather than as a mechanism feature of a market design, making full alienation unavailable for the term $R(t)$ permits, and the rent finding, sharpened by a directed falsifier of the labour remedy, making redistribution a pooling of bargaining position rather than an unattainable replication of scale. That conjunction, priced on a finite, adversary-relative, time-drifting reconstruction bound $R(t)$, is the paper’s defensible residue, and it is offered as a candidate that survived a named prior-art search, not as an empty cell.

Personal-data-market design and its failures. Spiekermann, Acquisti, Bohme and Hui (2015) catalogue why personal-data markets fail: asymmetric information, externalities, and adverse selection. Anderson (2001) and the WEIS tradition read these markets through Akerlof’s (1970) market for lemons: when a buyer cannot distinguish high-quality consented fresh data from noise, the market unravels. This is the literature the paper most directly answers on the design side. We take the field’s diagnosis and propose an architectural repair with three parts (Section 4): the structural separation whose guarantee is imported and time-indexed, a non-transferable Sybil-resistant quality stake, and a propose-and-respond bargaining protocol. The move we defend is not the observation that these markets fail but the specific mechanism proposed to make one clear.

The claim is a predicate claim, not an ordinal one. The substantive thesis is that behavioural sovereignty satisfies the five defining predicates of a capital stock, made testable in Section 3: returns, compounding (reputationally and as a conjecture; raw-data aggregation does not, Section 5.2), investability, opportunity access, and cross-context transfer. The thesis stands or falls on those predicates, not on a place in any enumeration. For provenance: the framing arose by analogy with integrated-reporting practice, which recognises six capitals (Financial, Manufactured, Intellectual, Human, Social and Relationship, and Natural, per the International Integrated Reporting Council); an earlier enumeration in the project’s prior lineage diverged from that list and is corrected here (Section 6). That analogy is the origin of the “seventh capital” label and nothing more; the reporting taxonomy carries no economic weight. In the project’s conjecture register the thesis is an occupied, architectural-class conjecture (C55); we state it as a thesis with a proof obligation, its five predicates being the content.

3. Model and preliminaries

We fix vocabulary and define every economic object before it is used.

Agents and the data subject. Let X be the data subject. Two agents act on her behalf or upon her. The *boundary agent* S observes and mediates: it is the surface through which the subject is watched and through which requests to disclose reach her. The *delegation agent* M acts: it carries out tasks the subject delegates. The architectural claim, developed in the companion paper and imported here, is that S and M can be separated so that S does not accumulate the task context M holds, and that context is erased between agent invocations (structural context erasure). The subject’s behavioural record is the object both agents could in principle reconstruct; the separation bounds what the observing side can. In the bargaining model of Section 4, “ M ” also names the single economic counterparty across

the table from X; the two uses coincide because the counterparty is the observing firm carrying both agents.

The appropriable surplus and the subject share. Normalise the appropriable per-subject, per-period *data-derived* surplus to 1. It is explicitly the data surplus, not the subject’s total welfare: the consumption value of any free service in a zero-price two-sided market is a separate object and is not in this pie (this scopes the claim and answers the “near zero *net of the service*” objection, which Section 5.1 carries as the upper allowance on the extraction share). The subject captures share $s \in [0, 1]$, the counterparty $1 - s$; the share is dimensionless, which is why the valuation can be stated in shares, gaps, and ratios without any monetary figure. We speak of two regimes of the one variable: an *extraction* regime, in which the interface drives s toward zero, and a *sovereign* regime, in which a bilateral protocol moves s into an interior range.

The realised-value gap. Let w_{gap} denote the ratio of the observer’s lifetime targeting-attributable capture per user to the one-time atomised broker-resale price of a record, both measured as stocks. It is large and positive. We argue in Section 5.2, on the returns-to-scale evidence, that this gap is market-position rent (an atomisation discount, a position rent, and lifetime accumulation) rather than super-additive aggregation value, and we give it an order-of-magnitude interval and decline a point value. A separate, genuinely data-value component, the single-record modelling premium w_{model} , is of order one and saturating, and is not super-additive: no external figure in the evidence set measures aggregation super-additivity, and the returns-to-scale record rejects it in the variety dimension (Section 5.2).

The market-maturity gate. The project’s value equation carries a multiplicative market-maturity factor $M(u, y)$ that gates realised value: no market able to price sovereignty, no value captured. In the source it has a name and a role but no functional form, domain, or units. We propose, as this paper’s own construction to be reviewed before it is treated as settled, $M : [0, 1] \times \mathbb{R}_+ \rightarrow [0, 1]$, with $u \in [0, 1]$ the market participation (the fraction of counterparties able to accept a 7012 proposal and settle against a subject-side aggregator) and $y \in \mathbb{R}_+$ market depth. A candidate form with the required boundary behaviour is

$$M(u, y) = u (1 - e^{-y/\tau}),$$

monotone non-decreasing, valued in $[0, 1]$, with $M(0, y) = 0$ and $M(u, y) \rightarrow u$ as depth grows ($\tau > 0$ a market-formation time constant). The *realised* share is $s_{\text{realised}} = M(u, y) s$: the bargaining share s of Section 4 gated by whether a counterparty able to price sovereignty is at the table. The form is a proposal; any form with the same boundary behaviour would serve, and τ is unestimated.

The capital properties as predicates. The thesis claims behavioural sovereignty has five properties of a capital form; we restate each as a checkable predicate rather than a metaphor. (i) *Returns*: holding the stock and disclosing from it selectively yields coordination the subject could not otherwise obtain (the return is trust-enabled access, not a coupon). (ii) *Compounding*: this holds in one sense and is rejected in another. A reputation built from prior consented disclosures may lower the cost of the next, so returns accrue on the accumulated relationship stock; this is a directional conjecture (Proposition P5, corresponding to a design assumption in the extraction), unmeasured, and carried as such. Aggregation of the *raw* behavioural record, by contrast, does not compound: the returns-to-scale evidence shows aggregate data value saturates and rejects super-additivity in the variety dimension (Section 5.2). The compounding the thesis relies on is reputational and conjectural, not a super-additivity of raw data, which the evidence denies. (iii) *Investability*: the separation architecture is the capital good in which the subject invests to protect the stock. (iv) *Opportunity access*:

the stock gates entry to coordination that requires demonstrated trust. (v) *Cross-context transfer*: a portable, subject-held record can be presented across counterparties. These are the content of the thesis, not demonstrated facts; Section 6 records which remain unmeasured.

The economic character, in one figure. The capital class the predicates describe is land-like, and the analogy is offered as an organising figure whose every leg is carried elsewhere in the paper as a result or a stated assumption, not as an argument in itself. Land is the canonical capital asset that yields rent by position rather than value by aggregation; one cannot make more of it by piling up more of the same, and the marginal product of one additional unit is near zero at scale, which is the returns-to-scale finding of Section 5.2. Its worth is locational, set by what the road connects it to: here the consent interface is the road, and the value is the appropriation share of Section 4. Its exposure erodes on a calendar with nothing added to the plot: the archive is fixed and the adversary’s background accumulates, the erosion clock of Assumption A3. And it is leased, never conveyed: a scoped disclosure transfers a season’s harvest, not the ground, because the observing side cannot reconstruct the underlying record from it for the term the erosion clock permits, which is the structural-inalienability answer of Section 2. The one quantity that genuinely accumulates is the cultivated fertility of the relationship, the reputational stock of predicate (ii), and it is a conjecture, not a measured aggregation of records. The figure adds no claim; it states the shape the five predicates and the imported guarantee jointly take.

Preconditions of the imported guarantee. The information-theoretic guarantee that underwrites the durability of the stock is imported from the companion paper. We state its preconditions here as numbered assumptions, so that later sections can reference them, and so that the guarantee is never invoked without them (GR-7 discipline).

- **Assumption A1 (non-collusion).** The boundary agent S and the delegation agent M do not collude to pool the context each holds; formally the inter-agent conditional independence $I(Y_S; Y_M \mid X) = 0$ holds with no third channel carrying the residue.
- **Assumption A2 (fixed adversary class).** The guarantee is stated against a specified adversary class of bounded capability; it is not a claim against every possible adversary.
- **Assumption A3 (time-indexing and the two deficit conditions).** The companion paper separates two clocks, and the economics prices the second. The first is certificational: declared capacities are certified against the estimation and attack tooling available at time t , and what drifts as tooling improves is the certified value, not the information. The second is the erosion clock. Let B_t denote the adversary’s accumulating background information (its linkage corpus and side priors, carried about the source through the world and not through the disclosed transcript); the residual uncertainty $H(X \mid B_t)$ is non-increasing in calendar time while $H(X)$, the capacities, and every certificate stay fixed. A1 and A2 license the additive capacity structure and an error floor but do not by themselves place the residual below one. The strict bound holds exactly while the informed capacity-deficit condition $C_S + C_M < H(X \mid B_t)$ holds, a measurable numerical fact about a system, an adversary class, and a declared background model, not a consequence of the architecture. The governing ratio is $R(t) = (C_S + C_M)/H(X \mid B_t)$, non-decreasing in t , with finite shelf life $t^* = \sup\{t : R(t) < 1\}$ coinciding with the first crossing. The erosion’s form and direction are proved in the companion for a declared background model; its rate is a register conjecture, not a theorem. What expires at t^* is the informed deficit condition, not the architecture.

The threat model of the economics is a passive observer that appropriates surplus. The imported guarantee prevents that observer from reconstructing, under A1 and A2 with the deficit condition

declared and for the duration $R(t)$ permits, the record whose sovereignty is being priced.

A synthesis to foreground: the separation’s incentive design underwrites A1. Assumption A1 is a statement about behaviour; Proposition P4 (Section 4) is one about incentives. Under two domain-scoped units of account with no exchange between them, P4 makes maintaining the separation the earnings-maximising strategy, so non-collusion is chosen by rational agents rather than merely hoped for. In this precise sense P4 is a sufficient condition for A1 to hold as an equilibrium: it *discharges* A1, and A1 only. It does not touch A2, the deficit condition, or the time-indexing of A3, and it holds only while the two token domains stay frictionally separate (Proposition P4 states the threshold and the honest reversion). Even with A1 fully discharged, $R(t)$ still drifts and t^* is still finite; the discharge secures non-collusion, not durability.

The quality stake. Reputation in the model is grounded in a non-transferable, non-purchasable proof-of-practice resource earned through sustained sovereign practice and spent on graph inscriptions. Its defining economic property is non-transferability: it cannot be minted by fake accounts, bought to bypass the practice requirement, or moved between accounts. We define it here as the quality-and-Sybil signal used in Section 4.3 and carry its Sybil-resistance property forward as a conjecture, not a theorem (register conjecture C42), with the proof obligation stated where it is used.

4. Results: economic propositions

We state the economic results as propositions, each with its preconditions. Where a proposition rests on a register conjecture we say so and give the proof obligation and the unstarted step; we attach no confidence band to any of them. The information-theoretic guarantee itself is not among these propositions: it is imported from the companion paper and cited.

Proposition P1 (the consent interface sets the appropriation share). Normalise the appropriate per-subject, per-period data surplus to 1, split between X (share s) and the counterparty M (share $1 - s$). Model the consent interface as the extensive form of one non-cooperative alternating-offers bargaining game (Rubinstein 1982): the interface fixes which party posts the first offer and how many offer/response rounds precede exercise of an outside option, while the pie and the outside options h_X, h_M are held fixed across interfaces. Then under a notice-and-consent interface $s \rightarrow 0$, and under a bilateral propose-and-respond interface s moves to an interior share equal to X’s bargaining power.

Basis and obligation (self-contained). We use one model, not two. Rubinstein alternating-offers is chosen because the consent interface *is* the extensive form of the game; the cooperative Nash bargaining solution enters only once, as the frictionless ($\Delta t \rightarrow 0$) limit of that game’s subgame-perfect equilibrium (Binmore, Rubinstein and Wolinsky 1986), and is used only to name the interior interval, never as a second, interchangeable model. A cooperative solution concept abstracts away the extensive form and so cannot distinguish the two interfaces except by an ad hoc bargaining-power asymmetry; the non-cooperative model derives the asymmetry from the protocol, which is the whole content of the proposition.

- *Extraction regime (notice-and-consent).* The interface gives the subject no power to propose: she sees posted terms and may accept or be excluded, a single-round ultimatum game with M as sole proposer. If X declines she realises her outside option h_X ; because she is not admitted to the aggregation or real-time-bidding venue where behavioural data clears, the only price

she can realise is the atomised broker-resale price of one record, so $h_X \approx 0$. The unique subgame-perfect share is $s_{\text{extraction}} = h_X \rightarrow 0$. This is *not* produced by the near-zero outside option alone: a symmetric two-sided bargain with both outside options near zero would split near one half. The extraction result requires *both* features the interface supplies, the single-round sole-proposer extensive form and exclusion from the clearing venue. This is the precise sense in which the interface, not the subject's preference, sets the share.

- *Sovereign regime (IEEE 7012 propose-and-respond)*. The standard inverts the offer direction: the subject posts machine-readable terms first, and the organisation accepts, counters once, or declines. This is an alternating-offers game with X as first proposer and genuine two-sidedness. A term set can be presented to more than one counterparty, and a subject-side aggregator gives an alternative route to realise value, so X's outside option is now credible; taken as roughly symmetric and non-binding, the outside options set threat points but do not pin the split, which is governed by bargaining powers (the outside-option principle; Binmore, Rubinstein and Wolinsky 1986; Sutton 1986). With X proposing first the subgame-perfect share is interior for any discount factors, and in the frictionless limit equals X's bargaining power $\beta = \rho_M / (\rho_X + \rho_M) \in (0, 1)$. The interior interval $[0.3, 0.7]$ of Section 5.1 is the image of β over a bounded, roughly symmetric range of relative patience (neither party more than about 2.3 times as impatient as the other): 0.5 symmetric, 0.3 counterparty-favourable, 0.7 subject-favourable. It is a bargaining projection with a stated precondition, not a measured or guessed band; no deployed sovereignty market has been observed to price s (Section 6), and we port no value of β .
- *Realisation*. The realised share is $s_{\text{realised}} = M(u, y) s$, making $[0.3, 0.7]$ the *mature-market ceiling*, approached only as participation and depth grow ($u \rightarrow 1, y \rightarrow \infty$), never promised. It also gives the extraction regime a second, market-structural reason for the near-zero share: today no counterparty prices sovereignty, so $u \approx 0, M \approx 0$, and the subject captures near-zero even of a share she might in principle win. Using $M(u, y)$ *weakens* the sovereign result in the correct direction; no value of M or τ is asserted.

Proposition P2 (the observer-to-broker gap is positive and is rent, not super-additive aggregation). The observer's lifetime targeting-attributable capture per user exceeds the atomised broker-resale price of a record by a large, positive, order-of-magnitude interval; that gap is market-position rent (an atomisation discount, a position rent, and lifetime accumulation), not super-additive aggregation value. *Basis and obligation*. The returns-to-scale evidence shows aggregate raw-data value saturates: forecast error falls only at a diminishing rate, the variety dimension is flat-to-negative so super-additivity in variety is affirmatively rejected, a small subsample suffices for a given estimation task, and combined-attribute broker profiling is barely above chance (Bajari, Chernozhukov, Hortacsu and Suzuki 2018; Varian 2014; Neumann, Tucker and Whitfield 2019, all external). We assert the sign, the order-of-magnitude interval, and the rent attribution, and we do *not* assert a point value or a super-additive adjective. The obligation is to resist letting the interval harden into a point and to resist re-importing super-additivity: the exact gap depends on region, tenure, and which broker price anchors the denominator, none of which the data identify. This is a result against the project's prior essays, which leaned on aggregation compounding, and Section 6 reports it as such.

Proposition P3 (a non-transferable stake resists the Sybil attack that unravels the market). A personal-data market fails when a buyer cannot distinguish high-quality consented data from noise; the failure is informational (Akerlof 1970; Spiekermann et al. 2015). A non-transferable, non-

purchasable proof-of-practice stake makes the cost of a quality claim equal to the difficulty of earning the stake, which resists the Sybil attack that would otherwise flood the signal with cheaply minted identities. *Basis and obligation.* This is a conjecture (register conjecture C42), carried with its proof obligation and no confidence band. The cost-to-forge argument is explicit: non-transferability closes the three cheap paths to a false signal (mint via fake accounts, purchase, transfer). The unstarted step is the adversary-regime analysis that would establish no cheaper path exists and bound the residual Sybil advantage; until it is discharged the proposition is a candidate repair for the lemons failure, not a demonstrated one.

Proposition P4 (the separation is incentive-compatible, up to a conversion- friction threshold). If the two agents earn in two domain-scoped units with sufficiently high conversion friction between them, protection-domain earnings buying only protection tooling and delegation-domain earnings buying only delegation tooling, then maintaining the separation of S and M is the earnings-maximising strategy, whereas a single unified unit would make context-pooling earnings-maximising. *Basis and obligation.* The obligation is to state the agents’ payoff functions and show separation is a best response under the two-unit design and defection under the one-unit design. Two cautions are load-bearing. First, the premise is not the binary “no exchange pool exists” but a threshold: let colluding generate additional joint surplus $\Delta > 0$ and let $\Phi \in [0, 1]$ be the fraction of it each agent realises in its own numeraire at the shadow exchange rate net of conversion friction; non-collusion is the best response exactly when $\Phi \Delta < \kappa$, the margin forgone by colluding. Under strict no-exchange ($\Phi = 0$) this holds trivially; as the shadow market becomes liquid ($\Phi \rightarrow 1$) it fails once $\Delta \geq \kappa$. We report the reversion with equal prominence: a sufficiently liquid shadow exchange rate reintroduces collusion, and then P4 does not discharge A1 and the non-collusion precondition reverts to a bare assumption. Bounding the conversion friction against the collusion surplus is an open item (Section 6). Second, the source material also contains a sentence asserting that under the single-unit design the reconstruction ceiling “fails” and “mathematical guarantees collapse”; stated without preconditions or time-indexing, it is not imported. This proposition asserts only incentive-compatibility up to the threshold, discharging A1 alone; it makes no static statement about reconstruction, and $R(t)$ still drifts.

Proposition P5 (reputation capital may appreciate while surveillance data depreciates). Reputation built from consistent consented disclosure is claimed to appreciate over time, while surveillance data depreciates under a freshness decay. This is the one sense in which the compounding predicate of Section 3 survives, and it survives only as a conjecture. *Basis and obligation.* This is a directional conjecture, unmeasured (it corresponds to E4-C07, a design assumption in the extraction, not a register theorem). The supporting argument, that reputation is hard to forge because it requires simultaneous behavioural consistency, correct personal-meaning derivation, and narrative continuity across time, is a design claim, not a proof; no measured forgery-cost curve is offered. Crucially, this reputational compounding is the *only* compounding the thesis relies on: raw behavioural-data aggregation does not compound (Proposition P2), so the thesis rests on this conjecture and the other four predicates, not on any super-additivity of raw data. The measurement that would settle it is named in Section 6.

5. Valuation methodology and evaluation

This section carries the valuation. Every number appears in the same table as its assumptions, measured quantities and projections are visibly separated, and no magnitude from the project’s prior

essay lineage is reproduced as a measured fact; the evaluation is an analysis of published measurements, not a new experiment.

Two per-person magnitudes circulate in the project’s prior essays, a present-day value gap and a larger accessible-volume gap under full capture, and neither survives a tier-A rebuild as a point estimate. A gap of the form (potential value)/(currently captured value) is ill-posed in the extraction regime, because the denominator tends to zero and a ratio to a near-zero base is a division artefact; the larger figure was already reframed in the prior text away from an arithmetic ratio, and a quantity its own source declares non-arithmetic cannot be re-imported as a valuation ratio. We replace the first with a share methodology and the second with a rent-attributed gap.

5.1 The surplus-appropriation share. We replace the retired present-day gap with the bounded, range-valued share s of Section 3. Both regimes are stated as dimensionless intervals on the same normalised per-period surplus, so the appropriation gap is the arithmetic difference of two stated intervals, not a free-standing claim.

Quantity	Regime interval (dimensionless)	Endpoint basis (external evidence, cited not printed)
s (extraction)	order 10^{-4} to 5×10^{-2}	Lower/anchor: s is (the subject’s per-period realisable value) over (the per-period appropriable surplus). The only price the subject can realise in a market she accesses is the atomised broker-resale price of one record; taken as an annual realisation and divided by the observer’s gross annual advertising flow per user, this is of order 10^{-4} (broker resale prices per record, Steel et al. 2013 / OECD 2013; per-user gross advertising flow, Meta 2023 Form 10-K; both external). The anchor moves with the unmeasured annual resale multiplicity of a record and is reported as an order, not a point. Upper 5×10^{-2} : an unmeasured generous allowance for uncaptured consumer surplus from the zero-price side of the two-sided market (Section 6), not a measurement.

Quantity	Regime interval (dimensionless)	Endpoint basis (external evidence, cited not printed)
s (sovereign; projection)	[0.3, 0.7]	Nash bargaining split of the same normalised surplus once a credible outside option exists: 0.5 symmetric, 0.3 counterparty-favourable, 0.7 subject-favourable, the image of the bargaining-power parameter β under bounded relative patience (Proposition P1). This is a bargaining projection, not a measurement; no deployed sovereignty market has been observed to price s (Section 6).
WTA/WTP endowment ratio (value-based)	$[\approx 1.2, \approx 4]$	Ratio of the elicited monetary valuations to accept disclosure versus to pay for protection (Acquisti, John and Loewenstein 2013, external). The lower/anchor ≈ 1.2 –1.4 is the value wedge in that experiment; the upper reach brackets larger endowment effects in the broader literature. A separate participation-proportion ratio of order 5 in the same experiment is a ratio of the shares of subjects choosing privacy under the two framings, not a ratio of valuations, and is cited as direction-only corroboration of the endowment effect, never as its magnitude.

The appropriation gap is now derived rather than asserted. It is the movement of s from the extraction interval to the sovereign interval:

$$\log_{10} s_{\text{sovereign}} - \log_{10} s_{\text{extraction}} \approx (-1) - (-4) = 3,$$

a share-shift of about three orders of magnitude at the anchors, widening to about four at the lower extraction endpoint and narrowing at the free-service allowance. The shift is a consequence of the two stated intervals, expressed as a change in a dimensionless share and never as a ratio to a near-zero base. The single figure that circulated in the project's prior essays is not reconstructed; what

survives is the interval-to-interval movement, which is why the endpoints, not a headline multiple, are the content of this subsection.

5.2 The realised-value gap. The prior draft carried a single wedge, positive and super-additive, of order 10^1 to 10^5 . That object does not survive: its top divided an annual flow by a one-time stock, and its super-additivity is asserted by no external measurement. We replace it with the two distinct objects the single wedge conflated, give each an order-of-magnitude interval derived from stated endpoints, refuse to point-identify either, and drop the word “super-additive” entirely.

Quantity	Interval (dimensionless)	Endpoint basis (external evidence, cited not printed)
w_{model} = single-record modelling/targeting premium: a targeted-segment record over a general record within the same broker market (like-for-like, stock over stock)	order $\approx 10^0$, saturating, not super-additive, not point-identified	The anchor is a cross-sectional price-level ratio between two record types for one subject's record (Steel et al. 2013 / OECD 2013, external), of order a few. It does not measure the returns to aggregating many subjects' data, and no external figure in the evidence set measures aggregation super-additivity. Returns-to-scale evidence bounds any aggregation term to order $\approx 10^0$: forecast error falls only at a diminishing rate, with genuine but diminishing gains in history length and the data-variety dimension robustly flat-to-negative, i.e. super-additivity in variety is rejected (Bajari, Chernozhukov, Hortacsu and Suzuki 2019, external); a small subsample suffices for a given estimation task, so the marginal analytic value of additional volume at scale is ≈ 0 (Varian 2014, external); combined-attribute broker profiling is barely above chance (Neumann, Tucker and Whitfield 2019, external).

Quantity	Interval (dimensionless)	Endpoint basis (external evidence, cited not printed)
w_{gap} = observer-to-broker realised-value gap: the observer's lifetime targeting-attributable capture per user over the one-time atomised broker-resale price of a record (like-for-like, stock over stock)	order 10^4 to 10^5 , not point-identified , and it is rent , not aggregation value	Both terms are stocks (lifetime accumulated observer capture over tenure, the annual gross flow scaled by the external targeting-attributable fraction; a one-time record price), so the ratio is dimensionless and the flow-versus-stock boundary the prior draft crossed is removed (Meta 2023 Form 10-K per-user gross flow; the targeting-attributable fraction from the online-advertising literature; Steel et al. 2013 / OECD 2013 broker prices; all external). The gap is large because the record is atomised and the subject is not admitted to the clearing venue, not because data value compounds: it decomposes into an atomisation discount, market-position rent, and lifetime accumulation, none of which is super-additive aggregation value. It moves with region, tenure, and which broker price anchors the denominator, and is reported as an order, not a point.

The gap is where the counterparty's surplus originates and is exactly the object a subject-side aggregator (Section 5.3) would need to capture. We assert its sign and its order-of-magnitude interval; we do *not* assert super-additivity, because no external figure in the evidence set measures it and the returns-to-scale record rejects it in the variety dimension. We assert no point value, and the obligation (Proposition P2, Section 6) is to resist letting the interval harden into one, because the exact gap fixes a region, a tenure, and a denominator that the available data do not identify, and because most of its size is rent rather than data value. This is the honest residue of the retired high-end intuition: the observer-to-broker gap is large, its lower and upper orders are stated, and it is correctly attributed to atomisation discount, market-position rent, and lifetime accumulation rather than to compounding aggregation; it is neither a per-person monetary amount nor the essay figure.

5.3 The market mechanism, instantiated. The value is lost in the extraction regime and realised in the sovereign one through three concrete instruments, and they are three *distinct* mechanisms,

not one.

- *Price discovery* is the consent interface read as a bargaining protocol (Proposition P1): notice-and-consent sets s near zero; IEEE 7012 propose-and-respond moves it into the interior range, gated to the realised share by market maturity $M(u, y)$.
- *Rent capture* (not wedge capture) is a subject-side aggregator. Today the observer-to-broker gap accrues entirely to the observer, but that gap is not super-additive aggregation value the aggregator would have to reproduce: the returns-to-scale evidence shows raw-data value saturates (Section 5.2), and the gap is market-position rent, the discount an atomised record suffers because the subject is not admitted to the venue where behavioural data clears. A data coalition, mediator of individual data, or data trust (Arrieta-Ibarra et al. 2018; Posner and Weyl 2018; Delacroix and Lawrence 2019) aggregates *bargaining position* on the subject side and returns a negotiated fraction of that rent to subjects. It redistributes a positional rent; it does not unlock latent compounding value, because the evidence says there is none to unlock. This is the honest and, we argue, stronger form of the claim: the remedy does not require individuals to replicate platform-scale aggregation.
- *Quality signalling* is the non-transferable stake (Proposition P3), the candidate repair for the lemons failure that would otherwise unravel the market of the first two instruments.

Stated plainly: in the extraction regime the share s is near zero and the observer-to-broker gap accrues entirely to the observer, so the sovereignty has a market price currently paid to someone else. The two levers are distinct: price discovery raises the subject's share of the pie she is admitted to; rent capture is about admission to the clearing venue in the first place. This is the home thesis as a mechanism claim, with the mechanisms named and kept distinct, and no monetary figure or reproduced ratio.

5.4 The tier ladder as illustrative parameters. The model sketches a reputation ladder in which higher accumulated relationship credentials unlock higher-value coordination. The specific value multipliers attached to the ladder in the source are illustrative design parameters with no empirical derivation; the source itself labels them theoretical projections. We report the compounding thesis, that privacy enables trust, trust enables higher-stakes delegation, and higher stakes generate higher returns that attract better opportunities, and we bind that “compounding” explicitly to *reputation* and carry it as the directional conjecture of Proposition P5, never to raw-data aggregation, which does not compound (Section 5.2). The multipliers are reported as illustrative, never as measured returns; no multiplier value is asserted as an observed rate.

5.5 The compression-as-assessment mechanism. The coordination economy is claimed viable by a compression protocol under which agents exchange compressed representations with expand-on-demand rather than full records, driving per-interaction cost toward near-zero while preserving privacy, and under which compression fidelity is a quantified assessment (in the Promise Theory sense) that the knowledge-transfer promise was kept. The extractable claim is the mechanism, compression fidelity read as a measurable assessment signal; the specific compression ratios in the source are not reproduced as facts.

5.6 An independent second-substrate derivation. An independent build of the same separation architecture on a different substrate kept the custody split and dropped the reputation score entirely, disclosing instead a signed, decomposable evidence graph verified offline against issuer identifiers, which never emits a tier, score, or ranking (register conjecture C95, the evidence-graph anti-score). We present this as an independent derivation, not a competitor. Its sharper economic reading is itself

conjectural: an emitted reputation scalar is a computed projection correlatable across contexts and accumulable by an adversary, whereas single-use selective disclosure emits the minimum for one decision. That the scoreless posture is tighter in leakage terms is a conjecture whose magnitude is unproven; it rhymes with the register’s existence-leak and reading-not-authority conjectures (C66) but is not a measured result, and the evidence-graph format travels with its W3C Verifiable Credentials citation.

6. Threats to validity

This section gives the negative results the prominence of the positive ones. It is not a coda; it is a result.

The rent finding runs against the project’s own prior essays, and we report it as such. The evidence rejects super-additive aggregation of raw behavioural data: the observer-to-broker gap is market-position rent, not compounding data value (Section 5.2, Proposition P2). The project’s prior high-end figure gestured at aggregation compounding; that gesture is denied by the returns-to-scale record and retired. We report this with the prominence of a positive result, and it strengthens the paper on two axes. The thin-market objection to propertization is now answered without an unmeasured super-additivity assumption (Section 2), and the valuation rests no load on a quantity the evidence denies; the surviving “bigness” is rent, measured in kind as an atomisation and a position discount even though its magnitude is non-identified. A claim that survives the deletion of its most convenient premise is stronger, not weaker. The only compounding the thesis retains is reputational and conjectural (Proposition P5).

The valuation is not point-identified. The share s is bounded and its extraction value is defensibly near zero, but its sovereign value is a bargaining projection, not a measurement: no deployed sovereignty market has been observed to price s . The realised-value gap w_{gap} is asserted only in sign, an order-of-magnitude interval, and its rent attribution; the genuine data-value component w_{model} is of order one and saturating; neither is point-identified, and the paper must not let either interval harden into a point.

The unit of analysis remains subject-indexed, and the relational critique is not fully answered. The normative collective-privacy literature engaged in Section 2 (Viljoen 2021; Taylor, Floridi and van der Sloot 2017; Fairfield and Engel 2015) argues that data relations create population-level interests no individual holder can represent. The framework internalises the social-data externality into the per-subject bound and adopts a collective remedy instrument, but it does not provide governance of the population-level interest itself; a valuation conducted per subject cannot claim to. That gap is left standing as the strand’s residual objection, not resolved.

The market-maturity gate is a proposal. The functional form for $M(u, y)$ in Section 3 is this paper’s own construction, chosen for its boundary behaviour, neither established in the source nor measured, with τ unestimated. It is presented so the value equation can be written down honestly, not because the form is known to be correct.

Two propositions rest on open conjectures. The Sybil-resistance of the non-transferable stake (Proposition P3, register conjecture C42) is an open conjecture with an unstarted adversary-regime step. The appreciation-versus- depreciation claim (Proposition P5) is a directional design conjecture, unmeasured, its unforgeability argument a design argument, not a forgery-cost measurement.

Neither is reported as settled.

The separation’s incentive compatibility rests on a conversion-friction threshold. Proposition P4 discharges Assumption A1 only up to a threshold: if a sufficiently liquid shadow exchange rate forms between the two domain-scoped units, context-pooling becomes earnings-maximising again and A1 reverts to a bare assumption. Bounding the conversion friction Φ against the collusion surplus Δ , or conceding A1 is assumed independently, is an open item.

The tier multipliers are illustrative. The reputation-ladder multipliers of Section 5.4 are design parameters, not measured returns.

A reference enumeration was corrected. The six-capital reference frame was stated incorrectly in the project’s prior whitepaper lineage and corrected against the International Integrated Reporting Council source (Section 2). The divergence is recorded; the predicate thesis is unaffected, since it rests no weight on the ordinal.

The imported guarantee has a finite, adversary-relative horizon, and its citability is submission-gating. The guarantee is not a static property. It holds against a fixed adversary class (Assumption A2) under non-collusion (Assumption A1), and only where the informed capacity-deficit condition $C_S + C_M < H(X \mid B_t)$ holds does $R(t) = (C_S + C_M)/H(X \mid B_t)$ satisfy $R(t) < 1$; that deficit condition is a measurable numerical fact about a system, an adversary class, and a declared background model, not a consequence of the architecture. $R(t)$ is bounded below one at evaluation and rises as the adversary’s background information accumulates against a fixed archive, shrinking $H(X \mid B_t)$ while the capacities and every certificate stay fixed, so the shelf life $t^* = \sup\{t : R(t) < 1\}$ is finite; the erosion’s form and direction are proved in the companion for a declared background model, and its rate is carried in the register as a conjecture, not a theorem. We deliberately do not claim a full “term structure” of dated maturities, which would require a schedule for $R(t)$ the register carries only as a conjectural upward drift; we claim no more than a finite, adversary-relative horizon. What expires at t^* is the deficit condition, not the architecture. We do not claim adversaries cannot reconstruct the record; we claim a time-indexed, precondition-bound bound on what they can, sourced to the companion paper. Two consequences: a valuation ignoring the horizon overstates the stock’s durability; and the entire durability leg is gated on WP-07 being a real, citable companion at submission. If the companion that proves the guarantee (with Assumptions A1 to A3 and the erosion result A3 imports) is not citable when this paper is submitted, the durability claim must be withdrawn to a stated open problem, because this paper imports that result and does not re-prove it. This gating is a submission blocker, not a polish item. The metric-agreement half of the gate is resolved: the companion’s erosion result is stated on the adversary’s accumulating background information, not on computational capability, so the object Section 2 distinguishes from the differential-privacy composition clock is the object the companion proves.

Reporting commitment. A measured result that runs against the predictions of this paper, for instance a deployed sovereignty market that fails to move s despite a working 7012 protocol and a functioning stake, or a realised-value gap that measures far outside the asserted interval, will be reported with the same prominence as a confirmation. The honesty of the negative result is the moat, not an embarrassment to be minimised.

7. Conclusion

The argument decomposes into four claims held together conditionally. First, behavioural data is subject-owned capital: a stock the subject can hold, invest in, compound (reputationally, as a conjecture; raw-data aggregation does not, Section 5.2), and transfer, rather than a labour flow or a firm-owned asset, and the two objections that sank naive propertization, alienability and thin markets, are answered by structure rather than doctrine. Second, the subject captures a near-zero share of the surplus that stock generates, and that share is not a preference but the subgame-perfect equilibrium of a bargaining game whose rules are the consent interface; a bilateral propose-and-respond protocol moves the interior share to the subject's bargaining power, gated to a realised share by market maturity. Third, the value is realised or lost through a mechanism with three distinct levers, price discovery through the consent interface, rent redistribution through a subject-side aggregator, and quality signalling through a non-transferable Sybil-resistant stake, so the sovereignty has a market price currently paid to someone else. Fourth, what makes the stock durable is architectural, an information-theoretic separation of the observing boundary agent from the acting delegation agent, imported from a companion paper, holding against a fixed adversary class under non-collusion with a declared capacity-deficit condition, and time-indexed so it holds only for a finite, adversary-relative horizon t^* .

None of the four claims is asserted beyond its evidence. The valuation is a share and a rent-attributed gap with ranges, not two magnitudes from the prior essays; the one place the evidence contradicted those essays, the rejection of raw-data compounding, is reported at the prominence of a positive result. The market mechanism is buildable but unbuilt. The guarantee is proved elsewhere, dated, and gated on the companion being citable at submission. The conclusion is a decomposition, not a triumph, and the open items are its real content:

- Bound the realised-value gap w_{gap} empirically, or more tightly than a wide order-of-magnitude interval, keeping its rent attribution.
- Estimate the market-maturity gate $M(u, y)$ and its time constant τ from an operating market, and test the proposed functional form and the realised-share prediction $s_{\text{realised}} = M(u, y) s$.
- Bound the conversion friction Φ against the collusion surplus Δ so that Proposition P4 discharges Assumption A1 in equilibrium, or concede A1 is assumed independently.
- Discharge the Sybil-resistance conjecture (C42) with the adversary-regime step, or refute it.
- Measure the reputation-appreciation claim (Proposition P5) against a forgery-cost curve.
- Observe a deployed sovereignty market and test whether a working 7012 protocol and a functioning stake actually move the subject's share s .

The paper closes in the decomposed, conditional, and dated form in which its claims are true. Behavioural data is subject-owned capital on five predicates; the architecture that would let a subject hold it is specifiable; the guarantee that would make it worth holding is imported, on a clock, and gated on a citable companion; and the price of the sovereignty it protects is, for now, collected by someone else.

References

(Handoff to A4: verify each against templates/submission/references/pv_v6.bib and add. IEEE 7012 already in extraction citations; confirm bib key. Confirm the predicate-led retitle. New bargaining and propertization and returns-to-scale citations flagged below are tier-A permitted external sources.)

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